2 Is Matter Around Us Pure?

The tangy flavour of fizzy soft drinks appeals to almost everyone. Carbon dioxide is an ingredient that gives the drink its sour flavour and bubbles. The tangy flavour is due to carbonic acid, which is produced when carbon dioxide gas (in low quantities) is dissolved in water, whereas the bubbles or effervescent character is caused by carbonation itself.

What's Inside

Topic(s)
- ✓ Topic-wise Notes
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TOPIC 1

MATTER

Learning Objectives

- Students will learn to classify the matter.
- Students will learn to differentiate between the types of mixtures.
- Students will understand different categories of solution.
- Students will learn how to list the properties and types of solution.
- Students will understand the various terms related to solutions.
- Students will learn how to make use of data to calculate concentration of a solution.
- Students will learn about the colloids based on dispersed phase and dispersion medium.

Learning Outcomes

- Students will be able to classify matter into pure substances and mixtures.
- Students will be able to differentiate between homogeneous and heterogeneous mixtures.
- Students will be able to categorise the solutions into different categories.
- Students will be able to determine the components present in a solution.
- Students will be able to describe the properties of solutions.
- Students will be able to define various terms.
- Students will be able to interpret the data to calculate the percentage of a solution.
- Students will be able to determine the dispersion medium and dispersed phase of colloids.

Real Life Application

When you think of any marble substance, what comes to your mind is the 7^{th} wonder of the world- Taj Mahal, which is completely made up of white marble.

A slab of marble has a variety of materials that were essentially pressed together by the pressure of the Earth (and heat). Despite being in the same location (a solid rock), these materials are not chemically or otherwise connected to one another.

The rock's components might be separated from one another by applying pressure and heat without requiring the destruction of chemicals or other types of bonding. Because of this, marble is a mixture of different types of matter.

Let us now study the topic 'matter' in detail.

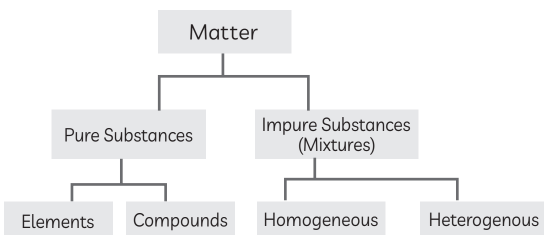
We observe different solutions like lemonade, cold drink, milk, etc., in our day–to–day life which are made of many substances. The food we eat and the detergent we use, are all a combination of different types of substances. These substances have different shapes, sizes, colours and nature. Thus, we can conclude that the matter around us is not pure and may contain impurities in them.

Anything that occupies space and has mass is called matter. It is divided into two types:

(1) Pure Substances

(2) Mixtures

Classification of Matter



Pure Substances

They are composed of only one kind of particles (atoms or molecules). They cannot be broken down into simpler forms by physical or chemical methods. *example* iron, copper, hydrogen, water, diamond, etc., are pure substances.

They have a fixed composition with fixed boiling and melting point.

Salt and sugar are pure substances

Mixtures

These are composed of two or more pure substances and contain two or more different kinds of particles (atoms or molecules). For examples: Sugar solution, salt solution, rocks, LPG, petroleum, etc. These can be separated by physical processes, like separation of salt from seawater by the process of evaporation.

Salt water is a mixture

Mixtures do not have a fixed composition or fixed melting and boiling point.

Types of mixtures

(1) Homogeneous Mixture

(2) Heterogeneous Mixture

OBJECTIVE Type Questions

[**1** mark]

Multiple Choice Questions

1. **Which of the following will show the "Tyndall effect"?**
   (a) Salt solution
   (b) Milk
   (c) Copper sulphate solution
   (d) Starch solution

Ans. *(b) Milk*

Explanation: Milk is a colloid that consists of fat and protein globules. When a beam of light is aimed towards a glass of milk, the light is scattered.

Related Theory

➥ *The intensity of scattered light is determined by the density of colloidal particles as well as the frequency of incident light.*

Assertion-Reason (A-R)

This question has two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

2. **Assertion (A): Alloys are a heterogeneous mixture of metal.**

   Reason (R): Alloys can be separated into their components using chemical methods.

Ans. *(d) A is false but R is true.*

Explanation: Alloys are homogenous mixtures of two or more metals in a fixed proportion. Thus, their separation is not possible by physical methods.

Related Theory

➥ *A homogeneous mixture of two or more metals or non-metals mixed in a molten state is known as an alloy.*

CASE BASED Questions **(CBQs)**

[**4** & **5** marks]

Read the following passages and answer the questions that follow:

3. *After studying the chapter 'Is matter Around us Pure', Yogesh, Ajay and Sandeep prepared mixtures using sugar, milk and chalk powder with water respectively. Teacher asked what observations they made.*

(A) Out of the three mixtures, which one is an example of suspension?

(B) Which mixture will show the Tyndall effect?

(C) What type of colloids are fog and smoke? Write the dispersed phase and dispersion medium in fog and smoke.

Ans. (A) Chalk dissolved in water is an example of suspension.

(B) (1) Since it becomes colloidal in nature, a mixture of water and milk exhibits the Tyndall effect.

(2) As chalk powder is insoluble in water, it will produce a non-homogeneous mixture. At first, the particles will scatter a beam of light, but after they settle down, they will no longer show the Tyndall effect.

(3) True solutions do not scatter light, and sugar solution is a true solution. As a result, the Tyndall effect does not appear in a sugar and water solution. *(Any two)*

Related Theory

➥ *When a powerful, converging beam of light is sent through a colloidal solution, the path of beam becomes visible at right angles to the beam. This is known as Tyndall effect.*

(C) Both Fog and Smoke are aerosols.

Colloid	Dispersed phase	Dispersion Medium
Fog	Liquid	Gas
Smoke	Solid	Gas

VERY SHORT ANSWER Type Questions (VSA)

[**1** mark]

4. **Sea water is a mixture of different salt compounds. One of these compounds is sodium chloride. Sodium chloride contains the elements sodium and chlorine. State the name of one other salt that does not contain either sodium or chlorine. [British Council 2022]**

Ans. Calcium carbonate.

Related Theory

➥ *Calcium carbonate is a chemical compound with the formula $CaCO_3$ formed by three main elements: carbon, oxygen, and calcium.*

SHORT ANSWER Type-I Questions (SA-I)

[**2** marks]

5. **Define a solute and solvent.[Diksha]**

Ans.Solute: The component of the solution that is dissolved in the solvent (usually present in the lesser quantity) is solute.

Solvent: The component of a solution that dissolves the other component in it (usually the component present in larger amounts) is called solvent.

SHORT ANSWER Type-II Questions (SA-II)

[**3** marks]

6. **Write the steps you would use for making tea. Use the words solution, solvent, solute, dissolve, soluble, insoluble, filtrate and residue. [Delhi Gov. SQP 2022]**

Ans. Below are the steps for making tea:

(1)At first, take 100 mL of water as solvent and boil it for few minutes.

(2)While solvent is boiling, add solutes, *i.e.,* milk, tea leaves, sugar.

(3)Now, boil the solution again for few minutes. Sugar gets dissolved as it is soluble in water.

(4) Tea leaves drops their colour into solution as filtrate. The remaining tea leaves being insoluble remains as residue.

(5) Now, filter the solution.

(6) Collect the filtrate in a cup.

LONG ANSWER Type Questions (LA)

[**4** & **5** marks]

7. **Richa investigated the solubility of three distinct compounds at various temperatures and compiled the following information (results are given in the following table, as grams of substance dissolved in 100 grams of water to form a saturated solution).**

Substance Dissolved	Temperature in K				
	283	293	313	333	353
	Solubility (in g)				
Potassium Nitrate	21	32	62	106	167
Sodium Chloride	36	36	36	37	37
Potassium Chloride	35	35	40	46	54
Ammonium chloride	24	37	41	55	66

(A)What effect does a change in temperature have on the solubility of salt?

(B)How much potassium nitrate would be required to make a saturated potassium nitrate solution in 50 grams of water at 313 K?

(C)Richa prepares a saturated potassium chloride solution in water at 353 K and lets it cool to ambient temperature. As the solution cools, what would she notice? Explain.

(D)Write the solubility of each salt at 293 degrees Fahrenheit. At this temperature, which salt has the highest solubility?

Ans. (A)Solubility increases with an increase in temperature.

(B) 62 g potassium nitrate dissolved in 100 g water at 313 K. So, to make a saturated potassium nitrate solution in 50 g of water, we will require the potassium nitrate.

$\frac{62}{100} \times 50 = 31$ g of potassium nitrate.

(C)Since the solubility of potassium chloride decreases, some soluble potassium chloride will crystallize and separate at room temperature.

(D) (1)Potassium nitrate has a solubility of 32 g at 293 K.

(2)At 293 K, sodium chloride has a solubility of 36 g.

(3) Potassium chloride has a solubility of 35 g at 293 K.

(4) Ammonium chloride has a solubility of 37 g at 293 K.

At this temperature, the solubility of ammonium chloride is highest.